

Transform-based particle filtering for state and parameter estimation in an Amari neural field model

by

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A Master's thesis

Submitted to the Department of Mathematics

Free University Amsterdam

July 2026

Date of Final Oral Exam: 23/06/2026, 14:00

Abstract

Neural field models are used to describe large-scale brain activity, but their parameters and underlying states are generally not directly observable. Recovering these quantities from noisy observations is therefore an important inverse problem in computational neuroscience. This thesis investigates joint state and parameter estimation in a one-population Amari neural field model using Bayesian particle filtering. The objective is to determine whether transform-based particle filtering, based on optimal transport, can match or outperform nested Sequential Importance Resampling (SIR). These methods are compared using synthetic observations generated from the Amari model. Parameter estimation accuracy and state reconstruction are evaluated across several observation noise levels and parameter regimes. The results show that both methods achieve comparable performance under all considered conditions. These findings suggest that, for the Amari model considered in this thesis, the additional computational complexity of transform-based particle filtering does not provide a practical advantage over nested SIR.

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1 Introduction

Neural field models are tools for describing large-scale brain dynamics [Coombes et al., 2014]. Among these, the Amari model [Amari, 1977] captures excitatory and inhibitory interactions across space, generating travelling waves, bumps and oscillations. The resulting activity patterns depend strongly on the parameters in the model. Considering that these parameters often have a physiological interpretation, such as the strength of excitation and inhibition, changes in their values may be associated with different dynamical regimes. Such regimes have been linked to different brain states, including sleep [Sanda et al., 2021]. Accurate estimation of these parameters is therefore important for understanding and characterizing observed neural activity.

However, neither these parameters nor the full underlying neural activity are directly observed, since only noisy and partial snapshots of the activity can be measured in practice. Consequently, both the underlying state of the system and the unknown parameters must be inferred from observations.

This leads to the inverse problem of inferring both the state and the unknown parameters from noisy and partial observations. Bayesian filtering methods are a natural approach to this problem, because they recursively update the posterior distribution as new observations become available [Reich and Cotter, 2015]. Since the posterior distribution is generally intractable, particle filters are used to approximate it using an ensemble of weighted particles.

A commonly used particle filtering method is sequential importance resampling [Gordon et al., 1993], which combines importance sampling with resampling to approximate the posterior distribution. However, transform-based particle filtering methods have demonstrated competitive or improved performance in a variety of applications [Al-Jarrah et al., 2023, Reich, 2013, Ruchi et al., 2019].

We therefore compare transform-based particle filtering with standard particle filtering, investigating whether it provides more accurate and robust state and parameter estimates for the Amari model using synthetic observations. In addition, we investigate different parameter regimes characterised by the relative strength of excitation and inhibition. Our results indicate that both approaches achieve comparable estimation accuracy across the considered regimes, with no consistent advantage of one method over the other.

The remainder of this thesis is organised as follows. Chapter 2 introduces the Amari neural field model, its discretisation and the data assimilation framework, including the generation of synthetic observations. Chapter 3 presents the Bayesian filtering framework and describes the particle filtering methods considered in this thesis. Chapter 4 reports the numerical experiments, evaluates the performance of the filtering methods across different parameter regimes, and discusses the limita-

tions of this study together with directions for future work. Finally, Chapter 5 summarises the main findings and concludes the thesis.

2 Model and data assimilation framework

2.1 Neural field model

Dynamical systems theory has long provided a framework for understanding patterns of neural activity. A classical example is the Hodgkin-Huxley model [Hodgkin and Huxley, 1952], which describes the generation of nerve impulses in single neurons and has motivated simplified reductions such as the FitzHugh-Nagumo model [FitzHugh, 1961]. At a larger scale, Beurle [Beurle, 1956] investigated signal propagation in randomly connected neural networks, laying the conceptual groundwork for later population-based approaches such as the Wilson-Cowan model [Wilson and Cowan, 1972], that describes coupled dynamics of interacting excitatory and inhibitory neural populations. Amari [Amari, 1977] extended population-based models to spatially continuous neural tissue by introducing one of the first neural field formulations, incorporating spatially structured synaptic interactions.

These ideas form the basis of neural field theory, which models large scale spatio-temporal neural dynamics by treating the neuronal populations as continuous media. In this framework, neural activity is described through spatio-temporal evolution equations that approximate the collective behaviour of large populations of interacting neurons.

In this work, we focus on the Amari model [Amari, 1977], which captures spatially structured synaptic interactions. We consider a one population Amari model describing the evolution of synaptic activity or voltage over a spatial domain $D \subset \mathbb{R}$, where D is a closed interval. The state variable $u(r, t) \in \mathbb{R}$ represents the synaptic activity at spatial location $r \in D$ and at time $t \geq 0$. These dynamics are governed by the integro-differential equation

$$\tau \frac{\partial u(r, t)}{\partial t} = -u(r, t) + \int_D w(r, s) \sigma(u(s, t)) ds + \Xi(r, t; \theta, a), \quad r \in D, \quad t \geq 0, \quad (1)$$

where $\tau > 0$ denotes the membrane time constant, which controls how fast the neural activity evolves over time.

Neural firing $\sigma(u(s, t))$ is modelled through a sigmoidal activation function

$$\sigma(u) = \frac{1}{1 + \exp(-\lambda(u - \mu))},$$

where λ controls the steepness of the firing rate and μ represents the activity threshold at which neurons become increasingly likely to fire.

The kernel $w(r, s)$ models spatial synaptic connectivity from point $r \in D$ to point $s \in D$, i.e. the strength of firing rate that $s \in D$ receives from $r \in D$. It is given by

$$w(r, s) = A_{ex} \exp\left(\frac{-\|r - s\|^2}{l_{ex}^2}\right) - A_{in} \exp\left(\frac{-\|r - s\|^2}{l_{in}^2}\right),$$

where A_{ex} and A_{in} are the strengths of the excitatory and inhibitory connectivity, respectively, while l_{ex} and l_{in} determine how far these connections extend spatially. We assume that there is short range excitation and long range inhibition, meaning $A_{ex} > A_{in}$ and $l_{ex} < l_{in}$. See Table 1.

The term $\Xi(r, t; \theta, a)$ represents an external forcing acting on the domain D . It is defined as

$$\Xi(r, t; \theta, a) = A \cos(2\pi(\nu r - (f + at)t)),$$

with A the amplitude, ν the spatial frequency and $\varphi(t) = f + at$ the time-dependent frequency. The parameter f is the initial temporal frequency and a determines the rate at which the temporal frequency changes over time.

This forcing is of particular interest because the time-varying frequency can significantly influence the resulting activity pattern. In particular, it can alter the speed and the structure of the emerging oscillations and the travelling wave solutions, and may induce transition between different dynamical regimes depending on the parameter values. Since oscillatory activity varies across brain states such as in sleep and wakefulness ([Krishnan et al., 2016, Sanda et al., 2021]), we therefore aim to infer the parameters governing this external forcing.

In this paper, the parameters of the connectivity kernel w , the firing rate σ , the membrane time constant τ and the factor a are assumed to be known and fixed, as seen in Table 1. The reasons for treating a as known are discussed in Chapter 4. Consequently, The focus will be on estimating θ in order to determine the external forcing acting on the cortical region D , while simultaneously estimating a discrete approximation of the state variable $u(\cdot, t)$ from noisy observations. The rest of this section introduces the discretised model used for data assimilation.

Parameter	Description	Value (unit)
τ	membrane time constant of neurons	0.03(s)
l_{ex}	spatial spread of excitatory connections (short range)	0.5(mm)
l_{in}	spatial spread of inhibitory connections (long range)	1 (mm)
A_{ex}	excitatory connectivity strength	10(mV)
A_{in}	inhibitory connectivity strength	6 (mV)
λ	steepness of the sigmoidal	5 (mV) ⁻¹
μ	activity threshold for neural firing	0.5(mV)

Table 1: Model parameters used throughout this paper unless stated otherwise.

2.1.1 Neural field model discretisation

Let $D = [0, L]$ and discretise the space using J evenly spaced nodes $r_j = j\Delta r$, $j = 0, \dots, J-1$, where $\Delta r = L/(J-1)$. The integral term in Equation (1) is approximated by using a composite trapezoidal rule, leading to the discrete interaction matrix $W \in \mathbb{R}^{J \times J}$ with entries

$$W_{ij} = w(r_i, r_j)b_j,$$

where $b_0 = b_{J-1} = \Delta r/2$, and $b_j = \Delta r$ otherwise. Let $U := u(\cdot, t) \in \mathbb{R}^J$ be the vector approximating $(u(r_0, t), \dots, u(r_{J-1}, t))^T$. Then we have the discrete spatial model

$$\tau \frac{dU(t)}{dt} = -U(t) + WS(U(t)) + I(t; \theta, a), \quad U(0) = U_0 \in \mathbb{R}^J. \quad (2)$$

Here $S(U(t)), I(t; \theta, a) \in \mathbb{R}^J$, with $S(U(t))$ approximating $(\sigma(u(r_0, t), \dots, \sigma(u(r_{J-1}, t)))^T$ and $I(t; \theta, a) = (\Xi(r_0, t; \theta, a), \dots, \Xi(r_{J-1}, t; \theta, a))^T$.

We discretize time by integrating Equation (2) on a temporal domain $[0, T]$, using a standard explicit Euler method [Butcher, 2016]. Therefore, for a given $k \in \mathbb{N}$, let $t_k = k\Delta t$ with $k = 0, \dots, K$ and let U_k be the approximating sequence of $U(t_k)$. Then we have

$$U_{k+1} = \Psi_k(U_k, \theta) := U_k + \frac{\Delta t}{\tau}(-U_k + WS(U_k) + I(t_k; \theta, a)), \quad U_0 = U(0). \quad (3)$$

The fully discrete formulation derived above provides the basis for the data assimilation procedure developed in the following sections. In particular, it enables the incorporation of noisy observations and the sequential updating of both the state variable and the unknown parameters.

2.2 Data assimilation

2.2.1 Data generation

To evaluate the data assimilation framework, we generate synthetic observations from the fully discrete Amari model introduced in the previous section. The underlying system is assumed to evolve deterministically according to this model, using the fixed parameter vector denoted by θ^* (see Table 2). Let U_0^* be the initial state on the spatio-temporal grid $[0, L] \times [0, T]$, with spatial and temporal discretisation levels J^* and K^* , respectively. Then the state evolves over time according to

$$U_{k+1}^* = \Psi(U_k^*, \theta^*),$$

where Ψ is the nonlinear update function defined in Equation (3). The observations follow the underlying system, however they are assumed to be noisy and partial. Therefore the high-resolution state $\{U_k^*\}$ is projected onto a coarser spatial and temporal grid with discretisation levels J and K respectively. This is achieved through an averaging procedure, resulting in a reduced representation denoted by $\{U_k\}_{k=0}^K \in \mathbb{R}^J$. In conclusion, the observations satisfy

$$Y_k = U_k + R\xi_k, \quad \xi_k \stackrel{i.i.d}{\sim} \mathcal{N}(0, \text{Id}_J), \quad k = 1, \dots, K,$$

where $R \in \mathbb{R}$ controls the magnitude of the observation noise. The procedure results in a dataset consisting of noisy, partial observations of the underlying deterministic system. The observations are represented in Figure 1. These observations serve as an input for the subsequent inference procedure.

Parameter	Description	Value (unit)
L	size of the cortical domain	10 (mm)
J	discretisation of the space $D = [0, L]$	25 (points)
J^*	discretisation of the space $D = [0, L]$ for the ground truth	500 (points)
T	time domain $[0, T]$	100 (s)
K	discretisation of the time domain T	10000 (points)
K^*	discretisation of the time domain T for the ground truth	20000 (points)
M	ensemble size of the particle set for the states	100 (states)
N	ensemble size of the particles set for the parameters	100 (parameters)
Q	noise from the state process	0.1
R	noise from the observation process	0.5
a	rate of change of temporal frequency	-0.005 Hz/s
θ^*	inferred parameter, with $\theta^* = (A^*, \nu^*, f^*)$	(1 mV, 0.1mm ⁻¹ , 0.5Hz)

Table 2: The values of the fixed parameters used in the data assimilation methods.

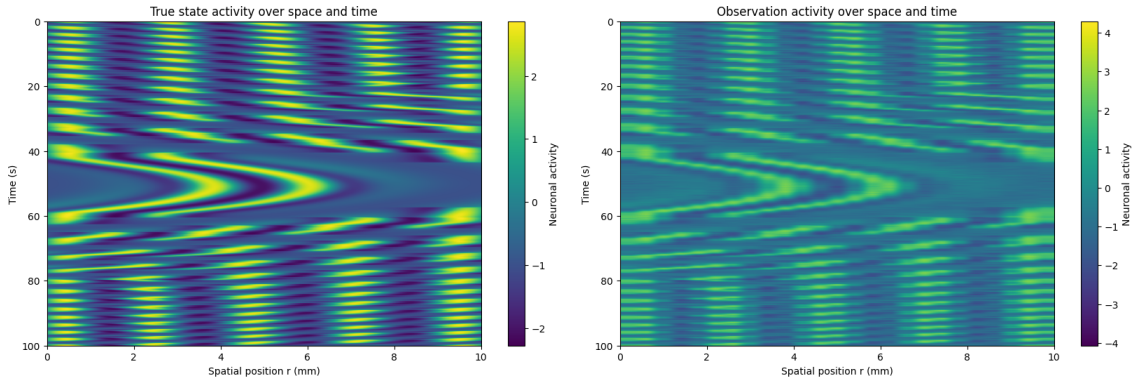


Figure 1: Generated truth ground state on the fine grid on the left and the noise partial observations on the coarser grid on the right with the parameters given from Table 1 and 2. The change of frequency is due to the factor $a = -0.005\text{Hz/s}$.

2.2.2 Inference setup

Given the synthetic observations described above, we now formulate the estimation problem within a data assimilation framework. The objective is to infer the unobserved system state and the unknown parameter θ , based on partial and noisy observations collected over time.

We model the hidden state of the system as a stochastic process $\{X_k\}_{k \geq 0}$, where each x_k is the realisation of the hidden state at time t_k . Likewise, y_k denotes the realisation of the observation process $\{Y_k\}_{k \geq 0}$. For $k \leq l$, we use the shorthand notation $x_{k:l} = \{x_k, \dots, x_l\}$ and $y_{k:l} = \{y_k, \dots, y_l\}$. We assume that the hidden state satisfies the Markov property,

$$\pi(x_k | x_{0:k-1}) = \pi(x_k | x_{k-1}).$$

Although the underlying Amari model is deterministic, we introduce process noise to account for model uncertainty and to facilitate effective data assimilation. The hidden state therefore evolves according to the stochastic state equation

$$X_k = \Psi(X_{k-1}, \theta) + Q\eta_k, \tag{4}$$

where Ψ is the non-linear model map defined in (3), θ is the vector of unknown parameters and $\{\eta_k\}$ are independent and identically distributed random variables with $\eta_k \sim \mathcal{N}(0, \text{Id}_J)$. The scalar Q controls the magnitude of the process noise.

The observations are assumed to depend only on the current hidden state,

$$\pi(y_k | x_{0:k}, y_{1:k-1}) = \pi(y_k | x_k).$$

Since the observations are generated by perturbing the hidden state with independent Gaussian noise, as described in Section 2.2.1, the corresponding likelihood is

$$\pi(y_k | x_k) = \mathcal{N}(y_k; x_k, R^2 I).$$

3 Bayesian sequential filtering methods

Bayesian filtering provides a framework for sequentially updating the distribution $\pi(x_k|y_{1:k})$ using incoming observations through Bayes' rule. The discrete Amari model introduced in Section 2.1 depends on an unknown parameter vector θ , which must be inferred from the observations together with the unobserved system state. This leads to the problem of sequential inference based on the joint posterior $\pi(x_k, \theta|y_{1:k})$.

In this thesis, the primary objective is the accurate estimation of the parameter posterior $\pi(\theta|y_{1:k})$. The state x_k is not of direct interest, but it is required to evaluate the likelihood of the observations under a given parameter value. Since this likelihood cannot be computed analytically, it must be approximated using particle-based methods.

To approximate the required likelihood, we first introduce standard particle filtering methods for state estimation under fixed parameters. In particular, we consider Sequential Importance Resampling (SIR) particle filtering [Gordon et al., 1993] and transform-based particle filtering [Cheng and Reich, 2013]. These methods will later serve as building blocks for nested formulations in which state filtering is used to evaluate the likelihood associated with different parameter values. The resulting nested methods are compared in Chapter 4 to evaluate their relative performance in parameter and state estimation.

3.1 Sequential importance resampling

The SIR particle filter is a Monte Carlo method used to approximate the posterior distribution $\pi(x_k|y_{1:k})$ via an ensemble of M particles and their corresponding weights. Given particles $\{x_{k-1}^m\}_{m=1}^M$, new particles are sampled according to the state transition equation (4). This yields samples $\{\tilde{x}_k^m\}_{m=1}^M$ from the predictive distribution $\pi(x_k|y_{1:k-1})$. The weights $\{w_{k-1}^m\}_{m=1}^M$ are updated using $\pi(y_k|\tilde{x}_k^m)$ and then normalised to obtain $\{\tilde{w}_k^m\}$. More generally, we have the update formula

$$\tilde{w}_k^m = \frac{w_{k-1}^m \pi(y_k|\tilde{x}_k^m)}{\sum_{i=1}^M w_{k-1}^i \pi(y_k|\tilde{x}_k^i)}. \quad (5)$$

A well known problem is degeneracy: after a few iterations, most particle weights become negligible except for a few particles with nearly all the mass. This happens because the likelihood $\pi(y_k|\tilde{x}_k^m)$ tends to be very peaked, so only particles close to the true state have reasonable weight values. Degeneracy is undesirable because a large ensemble effectively reduces to only a few representatives. Resampling mitigates this issue by discarding particles with very low weight and duplicating those with high weight, producing a new ensemble of M equally weighted particles.

A common diagnostic for particle degeneracy is the effective sample size (ESS), defined as

$$\text{ESS} = \frac{1}{\sum_{m=1}^M (w_k^m)^2},$$

which measure the number of effectively independent particles. When the ESS is low, most weight is concentrated on a few particles, indicating that the ensemble has collapsed.

In this work, resampling is performed at every time step. This is done to ensure a fair comparison between SIR and transform-based particle filter introduced in Section 3.2. Often, resampling is triggered only when the ESS falls below a threshold. This, however, can result in the two methods resampling at different frequencies, making it difficult to attribute differences in performance to the resampling mechanism itself. The ESS is still computed before resampling as a diagnostic to monitor particle diversity, but it is not a condition for resampling.

Following [Avitabile et al., 2025], stratified resampling is used to obtain a new set of equally weighted particles [Imbens and Lancaster, 1996]. We choose this method instead of standard multinomial resampling, because it yields a lower conditional variance. This means that for a fixed weighted particle ensemble, the resulting equally weighted particle set varies less from one resampling realisation than another, while retaining the same level of computational complexity ($O(M)$) [Imbens and Lancaster, 1996].

In stratified resampling, the cumulative sum of the normalised weights is divided into M equal intervals. In each interval, a single random number is drawn from a uniform distribution over that interval, and the particle whose cumulative weight interval contains this number is selected.

Even though resampling mitigates weight degeneracy, it can introduce a new problem known as sample impoverishment. If a few particles receive most of the weight, resampling duplicates these particles and while discarding others. As a result, the ensemble loses diversity, and after several filtering steps only a limited number of distinct particles remain. To restore diversity, a mutation step is applied using a Metropolis Hastings algorithm. The goal is to move the duplicated particles while preserving the target distribution $\pi(x_k|y_k, \theta)$. Given the resampled particles $\{\tilde{x}_k^m\}_{m=1}^M$, a new proposed state $\bar{x}_k^m \sim \pi(\tilde{x}_k^m|y_k, \theta)$ is acquired by propagating the resampled particles through state equation (4). The proposed state is accepted in the particle ensemble according to the Metropolis-Hastings acceptance probability

$$\alpha_m = \min \left(1, \frac{\pi(y_k|\bar{x}_k^m, \theta)}{\pi(y_k|\tilde{x}_k^m, \theta)} \right), \quad (6)$$

which compares the observation likelihood of the proposed particle with that of the current one. If the proposed particle explains the observations better, it is always accepted. Otherwise, it is accepted

with probability α . This procedure is repeated for a fixed number of times (mutation steps) for each particle. The complete mutation procedure is summarised in Algorithm 1, while the overall SIR particle filter is summarised in Algorithm 2.

Algorithm 1 Metropolis-Hastings Mutation

Let $\{\tilde{x}_k^m\}_{m=1}^M$ be the M resampled particles.
Set the number of mutation steps N_{MH} .
for $i = 1$ **to** N_{MH} **do**
 Propagate. Sample $\bar{x}_k^m \sim \pi(x_k \mid \tilde{x}_k^m)$ for $m = 1, \dots, M$ using state equation (4).
 Compute acceptance probability according to Equation (6).
 Accept/Reject.
 for $m = 1$ **to** M **do**
 Draw $u_m \sim \mathcal{U}(0, 1)$.
 if $u_m < \alpha_m$ **then**
 Set $\tilde{x}_k^m = \bar{x}_k^m$.
 end if
 end for
end for
Set $x_k^m = \tilde{x}_k^m$ and return $\{x_k^m\}_{m=1}^M$.

Algorithm 2 SIR Particle Filtering

Let $\{x_0^m\}_{m=1}^M \sim \pi(x_0)$ be the initial ensemble of M particles.
Set the initial weights $w_0^m = 1/M$ for $m = 1, \dots, M$.
for $k = 1$ **to** K **do**
 Propagate. Sample $\tilde{x}_k^m \sim \pi(x_k \mid x_{k-1}^m)$ for $m = 1, \dots, M$ using the state equation (4).
 Update weights according to Equation (5).
 Resample.
 for $m = 1$ **to** M **do**
 Draw $q_m \sim \mathcal{U}(\frac{m-1}{M}, \frac{m}{M})$.
 Select the unique index $p(m)$ satisfying $\sum_{j=1}^{p(m)-1} w_k^j < q_m \leq \sum_{j=1}^{p(m)} w_k^j$.
 Set $x_k^m = \tilde{x}_k^{p(m)}$ and $w_k^m = 1/M$.
 end for
 Mutate $\{x_k^m\}_{m=1}^M$ according to Algorithm 1
end for
Approximate the posterior by $\pi(x_k \mid y_{1:k}) \approx \frac{1}{M} \sum_{m=1}^M x_k^m$

3.2 Transform-based particle filtering

The SIR filter described in Section 3.1 uses stratified resampling to mitigate degeneracy. Stratified resampling minimises the variance of the resampling step among discrete resampling schemes [Imbens and Lancaster, 1996]. However, the resampling variance still remains, since for a given weighted particle ensemble, two independent resampling realisations will always produce different equally weighted ensembles. This variance propagates through the following time step, which could lower the accuracy of the final state estimate.

An alternative approach replaces the resampling step with a deterministic transport map that transforms the weighted particle ensemble into an equally weighted ensemble [Pele and Werman, 2009]. Because the transformation is deterministic, the method eliminates the resampling variance entirely. Moreover, rather than duplicating particles with high weights and discarding those with low weights, the transport map redistributes the entire particle ensemble in a continuous manner. Consequently, particle diversity is better preserved than stochastic resampling. The improved preservation of particle diversity has been reported to improve estimation accuracy in a range of high-dimensional and complex filtering problems [Al-Jarrah et al., 2023, Reich, 2013]. The remainder of this section describes this transformation in detail.

Let $\{\tilde{x}_k^m, \tilde{w}_k^m\}$ be the normalised weighted particles at time k . We want to have an equally weighted particle set $\{\hat{x}_k^m\}$ that approximates the same distribution. Instead of resampling, we compute an optimal transport map $T^* \in \mathbb{R}^{M \times M}$ that couples the source distribution, $\{\tilde{w}_k^m\}_{m=1}^M$, to the target distribution, the uniform distribution, while minimising the expected squared displacement. Formally, T^* solves

$$\min_T \sum_{i,j=1}^M T_{ij} C_{ij} = \min_T \sum_{i,j=1}^M T_{ij} \|x_k^i - x_k^j\|^2 \quad (7)$$

The objective function represents the total transport cost: the cost of transporting weight from particle x_k^i to x_k^j is proportional to the squared distance between them. The map is also subject to the following constraints

$$\sum_{i=1}^M T_{ij}^* = 1/M, \quad \sum_{j=1}^M T_{ij}^* = \tilde{w}_k^i. \quad (8)$$

The first constraint ensures each target particle j receives exactly $1/M$ total mass, while the second ensures source particle i distributes its weight $\tilde{w}_k^i$ across the target particles.

Finally, the new particles are obtained via the linear transformation

$$\hat{x}_k^j := M \sum_{i=1}^M T_{ij}^* x_k^i, \quad j = 1, \dots, M. \quad (9)$$

Since $\sum_{i=1}^M T_{ij}^* = 1/M$, we multiply by M so that the coefficients in equation (9) sum to one.

In practice, standard optimal transport can be computationally expensive: computing the exact transport map between two distributions of M particles has complexity $O(M^3 \log(M))$ [Pele and Werman, 2009]. To reduce computational cost, an entropic regularisation term is added. In particular, we use the Sinkhorn algorithm [Cuturi, 2013]. This turns (7) into a strictly convex problem

$$\min_T \sum_{i,j=1}^M T_{ij} C_{ij} + \epsilon \sum_{i,j=1}^M T_{ij} \log(T_{ij}),$$

with the constraints as defined in (8). The Sinkhorn algorithm solves this problem by rewriting the transport map as $T = \text{diag}(u_{\text{sink}}) K_{\text{sink}} \text{diag}(v_{\text{sink}})$. Here K_{sink} is a positive kernel matrix derived from the cost function with $K_{ij} = \exp(-C_{ij}/\epsilon)$, and u_{sink} and v_{sink} are scaling vectors. Instead of solving T directly, the algorithm iteratively updates u_{sink} and v_{sink} such that the resulting T satisfies constraints (8). Each operation consists of simple matrix-vector multiplications, and thereby reducing the complexity to $O(M^2 I_{\text{sink}})$ [Cuturi, 2013], where I_{sink} are the number of Sinkhorn iterations required for convergence. Depending on the convergence behaviour, this approach can be more efficient in practice than solving the exact transport problem. The quality of the approximation is controlled by the regularisation parameter ϵ : if $\epsilon \rightarrow 0$, the Sinkhorn solution converges to the classical optimal transport map, although very small values of ϵ may lead to numerical instability. In this work, we use $\epsilon = 0.1$ and compute the transport map using the Python Optimal Transport (POT) library [Flamary et al., 2021].

In one spatial dimension, however, the transport map has a simple construction based on the ordering of particles. Algorithm 3 illustrates this procedure. After sorting the particles in ascending order, probability mass is transferred sequentially from the weighted particles to the uniform target distribution. Starting from the lowest value, each source particle transfers mass to the current target particle. If the target weight reaches its limit, the source particle moves the next target; if the source particle has no weight left, we move to the next source. This procedure will be repeated until all mass has been assigned, yielding the transport matrix T^* . This works because, in one dimension, the optimal map is monotone and order-preserving.

Following the transport step, the transform-based particle filter applies the same Metropolis Hastings mutation as described Algorithm 1. Although the transport map preserves particle diversity more

Algorithm 3 One-dimensional transport map construction

Let $\{\tilde{x}_k^m\}_{m=1}^M$ be the weighted particles with associated weights $\{\tilde{w}_k^m\}_{m=1}^M$.
Set the target weights $W_j = 1/M$ for $j = 1, \dots, M$
Set the source index $i = M$ and the target index $j = M$
Initialise transport matrix $T = (T_{i,j}) \in \mathbb{R}^{M \times M}$
Sort. Relabel particles such that $\tilde{x}_k^1 \leq \tilde{x}_k^2 \leq \dots \leq \tilde{x}_k^M$
Permute the associated weights accordingly
Assign mass.
while $i > 0$ **and** $j > 0$ **do**
 if $\tilde{w}_k^i < W_j$ **then**
 $T_{i,j} = \tilde{w}_k^i$
 $W_j = W_j - \tilde{w}_k^i$
 $i = i - 1$
 else
 $T_{i,j} = W_j$
 $\tilde{w}_k^i = \tilde{w}_k^i - W_j$
 $j = j - 1$
 end if
end while
Return transport matrix T .

effectively than stratified resampling, the mutation step remains beneficial to maintain sufficient diversity over successive filtering steps.

3.3 Nested particle filtering methods

Nested Particle Filters (NPFs) approximate the joint posterior distribution $\pi(x_k, \theta | y_{1:k})$ using a two-layer particle filtering structure. The outer layer gives a particle approximation of the parameter distribution $\pi(\theta | y_{1:k})$, while for each parameter particle, an inner particle filter approximates the conditional state distribution $\pi(x_k | y_{1:k}, \theta)$ and provides an estimate of the likelihood $\pi(y_k | y_{1:k-1}, \theta)$. This likelihood estimate is then used to update the weights of the parameter particles. This framework extends both SIR and transform-based particle filtering to the joint state-parameter setting, which will be developed in this section.

Consider the parameter particle ensemble $\{\theta_k^n\}_{n=1}^N$ of size N and their corresponding weights $\{v_k^n\}_{n=1}^N$ at time step $k = 1, \dots, K$. For each parameter particle θ_k^n , there exists an inner state particle ensemble of size M , written as $\{x_k^{n,m}\}_{m=1}^M$ and their associated inner weights $\{w_k^{n,m}\}_{m=1}^M$. The parameter θ is assumed to be static, meaning it does not evolve according to a dynamical equation. However, the outer particles are jittered at each time step using the kernel

$$\kappa^N(\theta_k^n) = \left(1 - \frac{1}{\sqrt{N}}\right) \delta_{\theta_k^n} + \mathcal{TN}(\theta_k^n, N^{-3/2}, A_{\theta_k^n}, B_{\theta_k^n}), \quad (10)$$

which introduces controlled perturbations to maintain diversity in the parameter ensemble and pre-

vent parameter collapse of the outer particles onto a small number of parameter values. Here δ_{θ^n} is the point mass at θ^n and $\mathcal{TN}(\theta, N^{-3/2}, A_\theta, B_\theta)$ is the truncated normal distribution centred at θ^n with variance proportional to $N^{-3/2}$. The distribution is truncated to ensure that $A \in [0, 10]$, $\nu \in [0, 1]$ and $f \in [0, 1]$.

Let us consider the inner filter first. For each jittered outer particle $\tilde{\theta}_{k-1}^n$, the inner particle filter estimates the marginal likelihood $\pi(y_k | y_{1:k-1}, \tilde{\theta}_k^n)$. Starting from the previous inner ensemble $\{x_{k-1}^{n,m}\}_{m=1}^M$ with uniform weights $\{w_{k-1}^{n,m} = 1/M\}$, we first propagate the particles through the state equation (4) to obtain $x_k^{n,m} \sim \pi(x_k | x_{k-1}^{n,m}, \tilde{\theta}_k^n)$ for $m = 1, \dots, M$.

Once the propagated particles are obtained, the weights are updated using the likelihood

$$\omega_k^{n,m} = \tilde{w}_k^{n,m} \cdot \pi(y_k | \tilde{x}_k^{n,m}, \tilde{\theta}_k^n),$$

The marginal likelihood estimate for parameter $\tilde{\theta}_k^n$ is then computed as the average of the unnormalised weights:

$$L_k^n = \frac{1}{M} \sum_{m=1}^M \omega_k^{n,m}. \quad (11)$$

The weights are then normalised:

$$w_k^{n,m} = \frac{\omega_k^{n,m}}{\sum_{j=1}^M \omega_k^{n,j}}. \quad (12)$$

Resampling is performed at every time step to maintain a uniform distribution. The ESS is computed as a diagnostic to monitor particle diversity, but is not used as a condition for resampling. When using SIR, stratified resampling is used as described in Algorithm 2. When using transform-based particle filtering, optimal transport is used as described in Section 3.2. Finally, mutation is applied to reintroduce diversity, see Algorithm 1.

The output of the inner filter is the likelihood estimate $\hat{L}_k^n$ and the equally weighted particle ensemble $\{x_k^{n,m}\}_{m=1}^M$ for the use of the next time step.

Now we consider the outer layer. The outer weights are then updated as follows

$$\tilde{v}_k^n = \hat{L}_k^n v_{k-1}^n. \quad (13)$$

Finally, the outer weights are normalised to obtain $\{\hat{v}_k^n\}_{n=1}^N$. As with the inner loop, resampling is performed unconditionally at every time step. If SIR is used, stratified resampling is applied; if transform-based filtering is used, the optimal transport map is applied. Mutation is not applied

after resampling because the parameter is assumed to be fixed. Unlike the state particles, for which diversity must be maintained to represent the evolving state distribution, the parameter particles are intended to concentrate around a single fixed value as more data become available.

The output of the nested particle filter consists of an equally weighted parameter ensemble $\{\theta_k^n\}_{n=1}^N$ together with the corresponding equally weighted state ensembles $\{x_k^{n,m}\}_{m=1}^M$ for $n = 1, \dots, N$. A single state estimate can be obtained by first averaging over the inner state particles and subsequently averaging over the outer parameter particles. The corresponding parameter estimate is obtained by averaging over the equally weighted parameter ensemble.

The computational cost of the nested SIR method is $O(MN + N)$, while the nested transform-based method has cost $O(NM^2I_{\text{sink}} + N^2I_{\text{sink}})$. Since the inner layer dominates the runtime, the total cost is approximately $O(MN)$ for SIR and $O(NM^2I_{\text{sink}})$ for the transform-based approach. The details for nested transform-based particle filtering are described in Algorithm 4.

Algorithm 4 Nested Optimal Transport Particle Filtering

Let $\{\theta_0^n\}_{n=1}^N \sim \pi(\theta_0)$ be the initial parameter ensemble.
Set parameter weights $v_0^n = 1/N$ for $n = 1, \dots, N$.
for $n = 1$ **to** N **do**
 Sample $\{x_0^{n,m}\}_{m=1}^M \sim \pi(x_0)$.
 Set state weights $w_0^{n,m} = 1/M$ for $m = 1, \dots, M$.
end for
for $k = 1$ **to** K **do**
 for $n = 1$ **to** N **do**
 Parameter jitter. Sample $\hat{\theta}_k^n \sim \kappa^N(\theta_{k-1}^n)$.
 State propagation. Sample $\tilde{x}_k^{n,m} \sim \pi(x_k | x_{k-1}^{n,m}, \hat{\theta}_k^n)$ for $m = 1, \dots, M$.
 Weight update. Compute unnormalised weights $\omega_k^{n,m} = w_{k-1}^{n,m} \pi(y_k | \tilde{x}_k^{n,m}, \hat{\theta}_k^n)$.
 Normalise weights: $\tilde{w}_k^{n,m} = \omega_k^{n,m} / \left(\sum_{j=1}^M \omega_k^{n,j} \right)$.
 Marginal likelihood estimate. Compute $\hat{L}_k^n$ using $\{\omega_k^{n,m}\}_{m=1}^M$ according to Equation (11).
 Inner resampling (optimal transport). Apply Sinkhorn OT to $\{\tilde{x}_k^{n,m}, \tilde{w}_k^{n,m}\}_{m=1}^M$.
 Set $\{x_k^{n,m}\}_{m=1}^M$ equal to the resampled state particles
 Mutate $\{x_k^{n,m}\}_{m=1}^M$ according to Algorithm 1.
 end for
 Parameter weight update according to Equation (13).
 Normalise weights $\hat{v}_k^n = \tilde{v}_k^n / \left(\sum_{j=1}^N \tilde{v}_k^j \right)$.
 Outer resampling (optimal transport). Apply Sinkhorn OT to $\{\hat{\theta}_k^n, \hat{v}_k^n\}_{n=1}^N$.
 Set $\{\theta_k^n\}_{n=1}^N$ equal to the resampled parameter particles
end for
Approximate the parameter posterior by $\pi(\theta_k | y_{1:k}) \approx \frac{1}{N} \sum_{n=1}^N \theta_k^n$.

4 Numerical experiments

This chapter compares the nested SIR filter with the nested optimal transport (OT) filter for joint estimation of the state and the parameter θ . In the remainder of this chapter, the methods are referred to as SIR and OT, respectively. Unless stated otherwise, all experiments use the model and the parameters stated in Chapter 2.

Code Availability

The code developed for this thesis is publicly available at:

https://github.com/Nancy-Ntawumenyumunsi/Thesis_code_files.

4.1 Experimental setup

For each experimental configuration, 15 independent runs are performed for both filtering methods. Results are averaged over these runs unless stated otherwise. The number of mutation steps is fixed at five throughout all experiments.

The ground truth initial condition is defined on the spatial grid J^* by

$$U_0^*(r^*) = \sin(\pi r^*).$$

The filtering methods operate on the coarse grid J . The initial particle ensemble is sampled according to

$$X_0 \sim \mathcal{N}(\sin(\pi r), C), \quad C_{ij} = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\|r_i - r_j\|}{2L^2}\right).$$

The initial ensemble is centred around the true initial condition in order to focus the comparison on the assimilation process rather than on recovery from large initialisation errors.

The state estimation accuracy is assessed through the root mean squared error (RMSE) over time and the final state RMSE. Since the filters operate on the coarse grid J , the RMSE is computed using the interpolated ground truth and the estimated state. Parameter estimation performance is evaluated using the final parameter estimate. In addition, the variability of the final parameter estimates across runs is also reported. Finally, the ESS of both the parameters and the state are monitored at each time step. Although resampling is performed unconditionally, the ESS before resampling indicates how sharply peaked the likelihood is and how much diversity is lost prior to resampling. This diagnostic helps distinguish between configurations where poor performance arises

from weight degeneracy (low ESS) and those where it arises from other sources, such as changes in the underlying dynamics.

To investigate robustness of the filtering methods, several experimental scenarios are considered. The first corresponds to the baseline parameter settings introduced in Chapter 2. The second reduces the observation noise by setting $R = 0.1$. The third further reduces the observation noise to $R = 0.0001$. In the fourth configuration, the excitatory strength is decreased to $A_{\text{ex}} = 6$, making it equal to the inhibitory strength A_{in} . In this balanced regime, the synaptic kernel satisfies $w(r, r) = A_{\text{ex}} - A_{\text{in}} = 0$. This means that neurons no excite themselves. This changes the underlying neural dynamics and provides an additional test case for evaluating the robustness of the filtering methods. The final configuration combines the reduced observational noise, $R = 0.1$, and the reduced excitatory strength, $A_{\text{ex}} = 6$. Together, these scenarios assess filter performance under varying levels of observation noise and under modified neural dynamics.

4.2 Results

Before examining individual configurations, we note that the initial temporal frequency f does not converge to the true value for either method across all configurations, even though the mean appears close (See Figure 3 for the baseline configuration). This observation is discussed further in Section 4.3. We therefore focus on A and ν .

Baseline: We first consider the baseline configuration with $A_{\text{ex}} = 10$ and observational noise $R = 0.5$. Figure 2 shows the convergence of the parameter estimates over time for both SIR and OT. For SIR, the parameters A and ν converge to values close to the truth, 1.009 ± 0.063 for A and 0.092 ± 0.005 for ν (see Table 3). In particular, ν converges rapidly and remains stable after approximately 60 seconds, with a narrow uncertainty bandwidth indicating consistent estimates across runs. OT also converges to reasonable estimates, 0.991 ± 0.267 for A and 0.118 ± 0.106 for ν , but exhibits wider uncertainty bands. Inspection of the individual runs shows that this is mainly caused by a single run in which A and ν did not fully converge (see Figure 3).

Figure 4 shows the state estimation RMSE over time. The overall RMSE, final RMSE value and the time averaged RMSE are reported in Table 4. SIR achieves slightly lower RMSE than OT across all time steps: overall RMSE is 0.42 for SIR versus 0.50 for OT, and the time-averaged RMSE is 0.40 versus 0.47. The final RMSE is also lower for SIR (0.37 vs 0.44). However, given the overlap in uncertainty bands, these differences are likely not statistically significant.

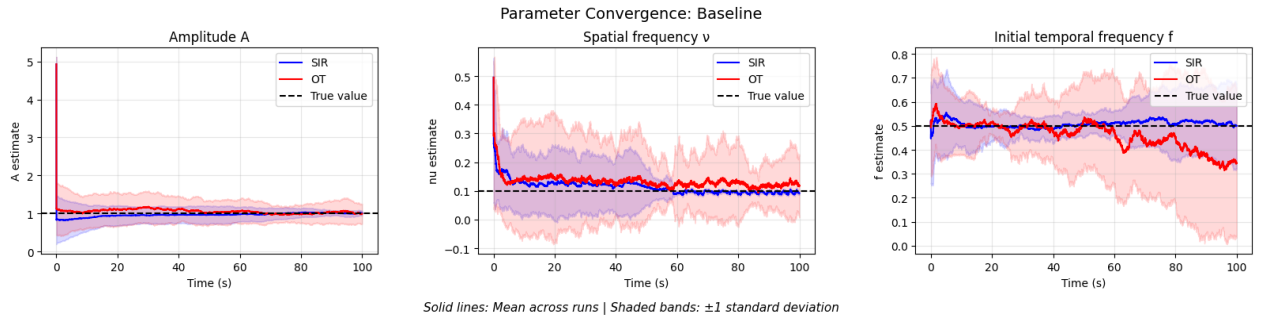


Figure 2: Parameter convergence in the baseline configuration. Solid lines show the mean estimate across 15 runs. Shaded bands indicate ± 1 standard deviation. The dashed lines indicate the true parameter values.

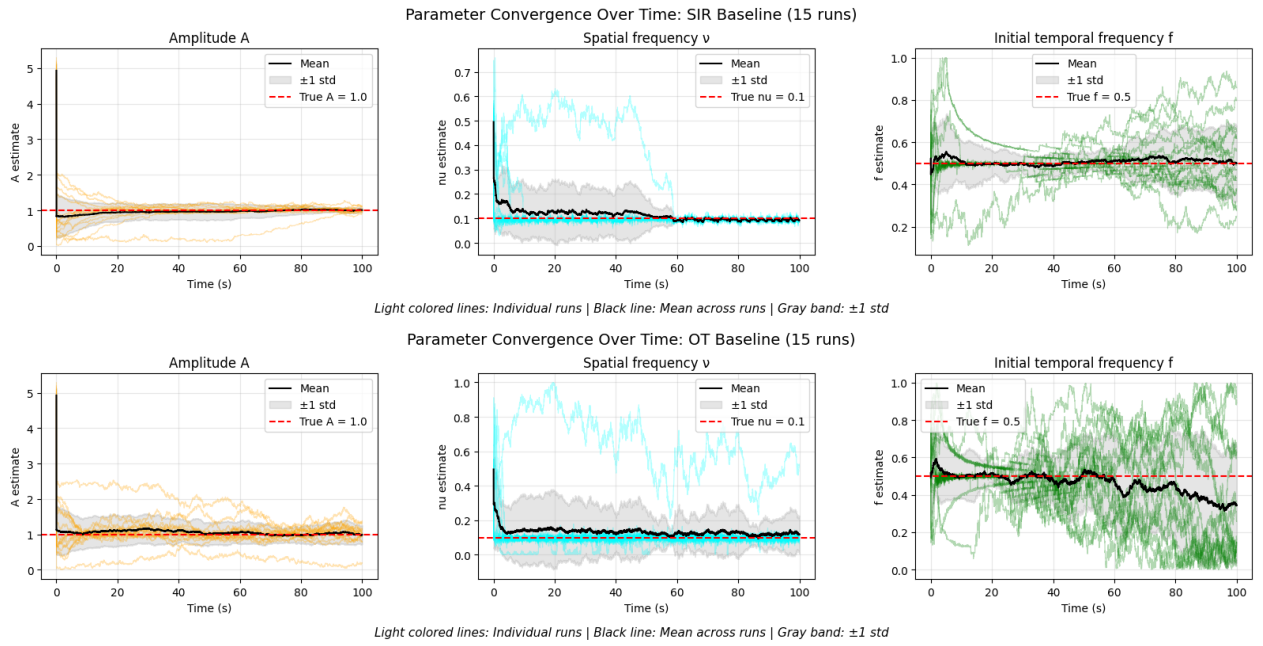


Figure 3: Individual parameter trajectories for (top) SIR and (bottom) OT in the baseline configuration. Each thin line represents one run; dashed lines indicate true values.

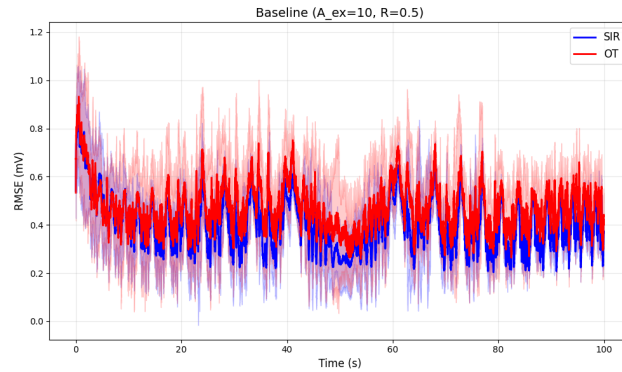


Figure 4: State estimation RMSE over time. The solid lines indicate the mean RMSE at that given time. Shaded bands indicate ± 1 standard deviation.

Low noise: We now reduce the observation noise to $R = 0.1$, while keeping $A_{\text{ex}} = 10$ fixed. For SIR, most runs still recover the parameters accurately, as shown in Figure 5. However, the variability of the estimates increases compared to the baseline, in particular for ν , which shifts from 0.092 ± 0.005 to 0.141 ± 0.191 . This increase in uncertainty is primarily driven by a small number of runs in which the filter fails to fully converge.

To understand this behaviour, we consider the effective sample size (ESS). As reported in Table 5, the mean parameter ESS drops from 52 in the baseline to 2 in the low noise configuration. This indicates that the filter operates in a more degenerate regime, where the importance weights become more concentrated after each time step. This effect is expected, as reducing the observations noise makes the likelihood $\pi(y_k | \tilde{x}_k^{m,n}, \hat{\theta}_k^n)$ more peaked, thus increasing weight variability. In the nested setting, this weight degeneracy affects the quality of the inner filter’s likelihood estimate $\hat{L}_k^n$, which can lead to occasional instability in the parameter update.

In contrast, OT shows little change compared to the baseline. The parameter estimates remain stable, with similar mean and variances across both configurations.

Figure 6 shows the state estimation RMSE over time. Both filters achieve nearly identical RMSE values (See Table 4), with overall RMSE for SIR and OT being 0.39. The decrease in RMSE under lower observation noise is expected due to more informative observations.

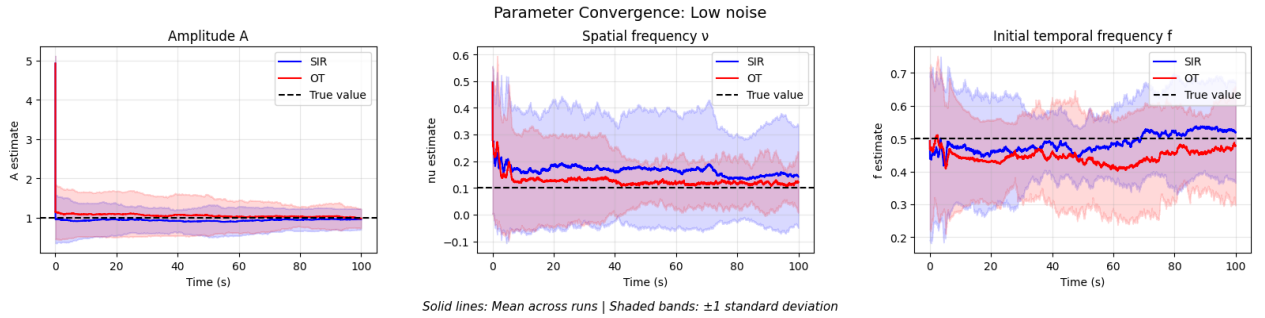


Figure 5: Parameter convergence for low noise configuration.

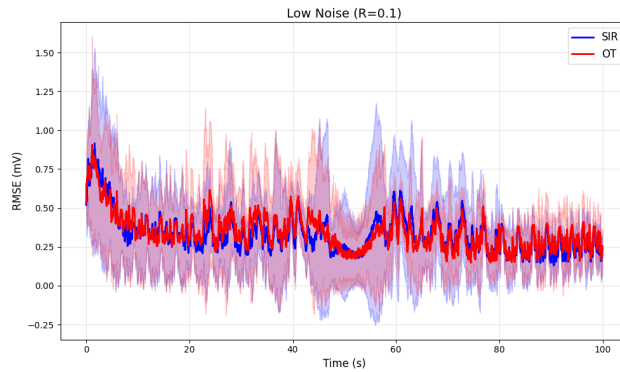


Figure 6: State estimation RMSE over time for the low noise configuration.

Very low noise: We further reduce the observation noise to the extreme value $R = 0.0001$, where observations are nearly noise-free. Figure 7 shows the parameter convergence for both methods.

For SIR, A converges well to the truth (1.110 ± 0.194). The estimate of ν remains close to the baseline in terms of its main value. However, increased variability is observed across runs, with occasional divergence after approximately 50 seconds. This behaviour is reflected in the large uncertainty estimate (0.145 ± 0.202). This instability can be explained by the combination of a highly peaked likelihood and the time-varying frequency in the external forcing: once the particles that could track the changing frequency are discarded, the mutation steps are insufficient to reintroduce them, leading to occasional loss of tracking ν .

For OT, the estimate of A remains close to the truth (1.092 ± 0.583), although the uncertainty increases compared to the previous configurations (± 0.267 in baseline and 0.243 in low noise). The estimate ν also increases compared to the baseline and low noise case, rising to 0.162 ± 0.200 , but remains within the same order of magnitude as in the other configurations. OT thus shows a moderate increase in variance, but most runs still converge to reasonable estimates.

Figure 6 shows the state estimation RMSE over time. At early times, SIR performs slightly better, but this difference diminishes over time. The resulting RMSE values are comparable between the methods (0.34 for SIR vs 0.52 for OT) while the final RMSE values are close (0.27 vs 0.32).

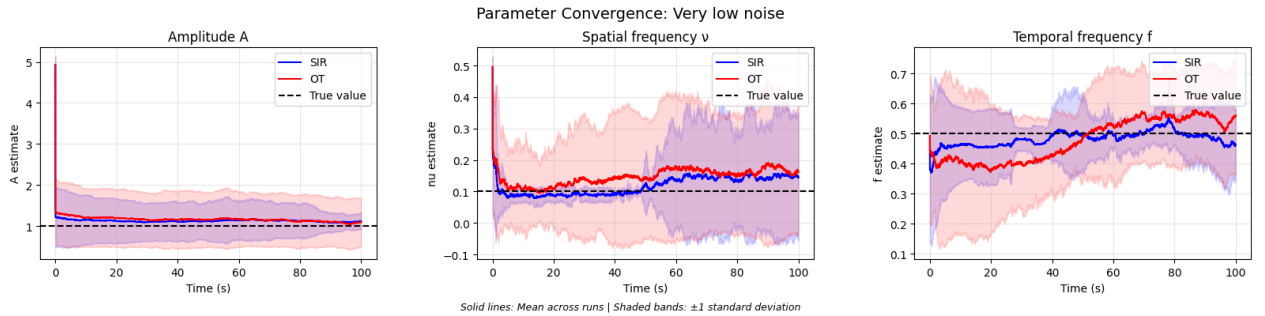


Figure 7: Parameter convergence for the very low noise configuration.

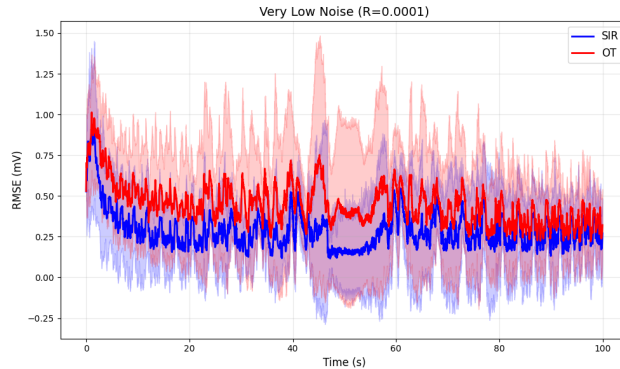


Figure 8: State estimation RMSE over time for the very low noise configuration.

Balanced kernel Next we turn to the balanced kernel configuration, where the excitatory and inhibitory strengths are equal ($A_{\text{ex}} = A_{\text{in}} = 6$). In this regime, the synaptic kernel satisfies $w(r, r) = 0$, meaning that neurons no longer excite themselves. To illustrate the resulting dynamics, Figure 11 shows a representative state trajectory for a single run. Figure 9 shows the parameter convergence over time for both methods.

For SIR, the parameter A remains close to the true value (0.952 ± 0.275). The parameter ν shows a positive bias (0.135 ± 0.105) and increased variability compared to the baseline configuration.

OT shows improved parameter estimates in this regime. Both A and ν converge to values close to the truth with narrower uncertainty bands: $A = 0.966 \pm 0.121$ and $\nu = 0.093 \pm 0.010$. The parameters typically converge shortly after 60 seconds, indicating consistent estimates across runs.

It is worth noting that the ESS is higher in this configuration (75 vs 50 in the baseline, see Table 5). This indicates better particle diversity before resampling, which may contribute to the improved performance of OT in this regime.

Despite OT's modest improvement in parameter estimation, state tracking performance remains comparable between the two methods. The overall RMSE is 0.19 for SIR versus 0.22 for OT, the time averaged RMSE 0.18 versus 0.20 and final time RMSE is 0.17 for both. SIR performs slightly better at early times, but the difference diminishes over time.

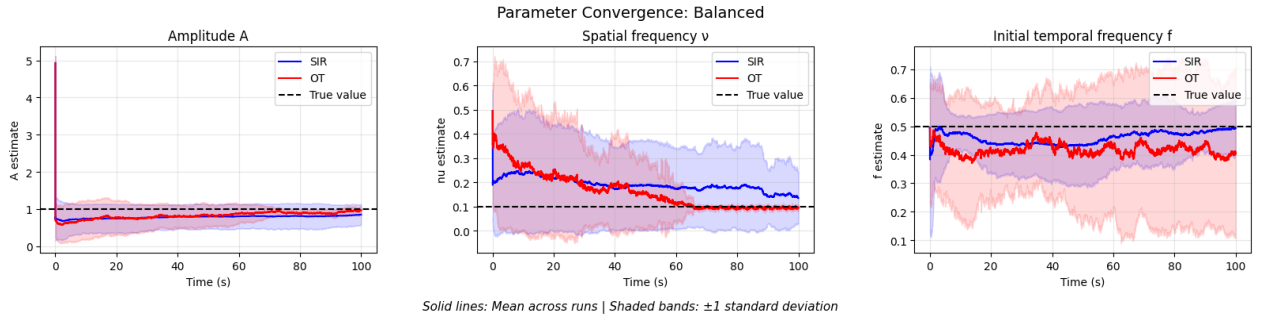


Figure 9: Parameter convergence for equal excitatory and inhibitory strengths.

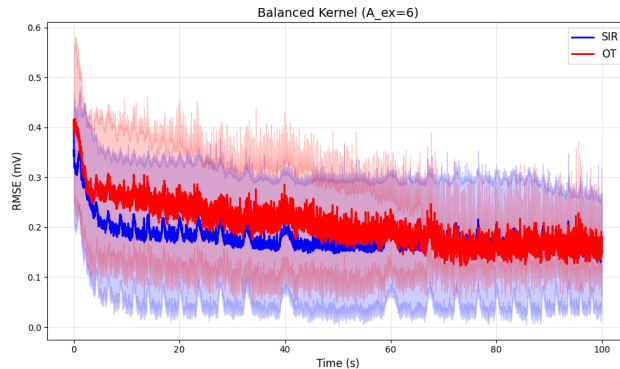


Figure 10: State estimation RMSE over time for $A_{\text{ex}} = 6$ configuration.

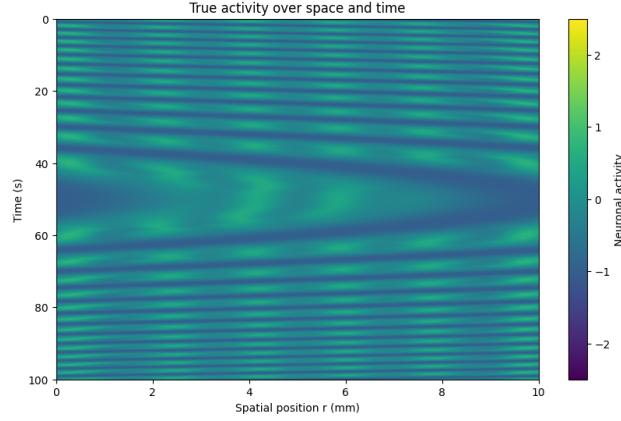


Figure 11: True spatio-temporal state in the balanced kernel configuration ($A_{\text{ex}} = 6$). Compared to the baseline (Figure 1 on the left), the activity is more muted and less sharp due to the absence self-excitation. At a fixed time, horizontal slices reveal more frequent spatial bumps, indicating more structured dynamics.

Combined: Finally, we consider the combined configuration, which couples the balanced kernel ($A_{\text{ex}} = 6$) with low observation noise ($R = 0.1$). Figure 12 shows the parameter convergence over time.

For both methods, the parameter A tends to underestimate the true value: SIR gives 0.868 ± 0.266 and OT gives 0.864 ± 0.293 . The parameter ν shows an upward shift: 0.143 ± 0.127 for SIR and 0.136 ± 0.161 for OT. The value of ν is comparable to the low noise and balanced kernel configuration, but A is noticeably lower. This suggests that A may be more affected by the combined change in noise level and kernel structure than ν , although further analysis would be need to confirm this.

Despite the shift in parameter estimates, the state estimation RMSE is the lowest of all configurations: overall RMSE is 0.12 for SIR versus 0.14 for OT, time-averaged RMSE is 0.11 versus 0.12, and the final RMSE is 0.10 for both methods. This is consistent with the improved state tracking observed in both the low noise and balanced kernel configuration.

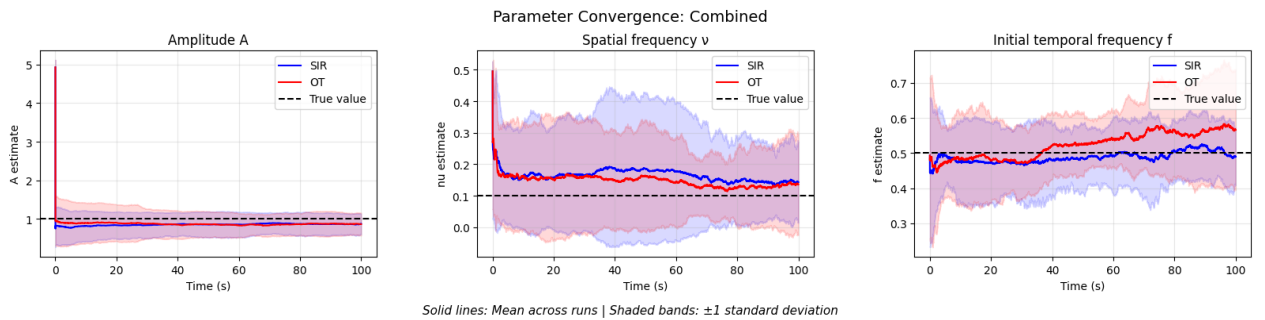


Figure 12: Parameter convergence for combined configuration ($A_{\text{ex}} = 6, R = 0.1$).

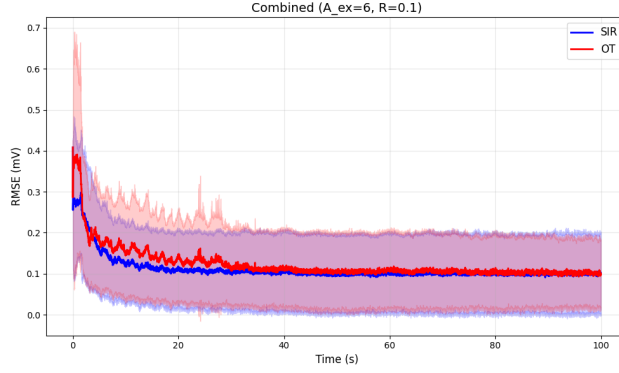


Figure 13: Estimated state RMSE over time for combined configuration.

Baseline ($A_{\text{ex}} = 10, R = 0.5$)			
Method	A	ν	f
SIR	1.009 ± 0.063	0.092 ± 0.005	0.500 ± 0.117
OT	0.991 ± 0.267	0.118 ± 0.106	0.344 ± 0.303
Low noise ($R = 0.1$)			
Method	A	ν	f
SIR	0.958 ± 0.266	0.141 ± 0.191	0.519 ± 0.153
OT	0.980 ± 0.243	0.122 ± 0.114	0.478 ± 0.177
Very low noise ($R = 0.0001$)			
Method	A	ν	f
SIR	1.110 ± 0.194	0.145 ± 0.202	0.163 ± 0.118
OT	1.092 ± 0.583	0.162 ± 0.200	0.557 ± 0.192
Balanced kernel ($A_{\text{ex}} = 6$)			
Method	A	ν	f
SIR	0.952 ± 0.275	0.135 ± 0.105	0.493 ± 0.096
OT	0.966 ± 0.121	0.093 ± 0.010	0.403 ± 0.297
Combined ($A_{\text{ex}} = 6, R = 0.1$)			
Method	A	ν	f
SIR	0.868 ± 0.266	0.143 ± 0.127	0.490 ± 0.080
OT	0.864 ± 0.293	0.136 ± 0.161	0.566 ± 0.170

Table 3: Final parameter estimates for all configurations.

4.3 Discussion

The aim of this work was to determine whether nested transform-based particle filtering improves the joint state and parameter estimation compared to nested SIR for the Amari neural field model. Across all five experimental configurations, both methods achieve comparable state reconstruction errors and parameter estimation accuracy, with only minor regime-dependent differences. The theoretical advantage of OT lies in preserving particle diversity through a transport map. However, this did not translate to consistently better estimates in our experiments. The lack of improvement may be due to the Amari model not being sufficiently complex or high-dimensional for the advantages of OT to become apparent under the tested conditions.

A consistent finding across all configurations was the non-identifiability of the initial temporal frequency f . The mean estimate can appear close to the true value, but individual runs often diverge,

Baseline ($A_{\text{ex}} = 10, R = 0.5$)			
Method	Overall RMSE	Final-time RMSE	Time averaged RMSE
SIR	0.42	0.37	0.40
OT	0.50	0.44	0.47
Low noise ($R = 0.1$)			
Method	Overall RMSE	Final-time RMSE	Time averaged RMSE
SIR	0.39	0.24	0.33
OT	0.39	0.26	0.34
Very low noise ($R = 0.0001$)			
Method	Overall RMSE	Final-time RMSE	Time averaged RMSE
SIR	0.34	0.27	0.29
OT	0.52	0.32	0.45
Balanced kernel ($A_{\text{ex}} = 6$)			
Method	Overall RMSE	Final-time RMSE	Time averaged RMSE
SIR	0.19	0.17	0.18
OT	0.22	0.17	0.20
Combined ($A_{\text{ex}} = 6, R = 0.1$)			
Method	Overall RMSE	Final-time RMSE	Time averaged RMSE
SIR	0.12	0.10	0.11
OT	0.14	0.10	0.12

Table 4: Overall RMSE, The final RMSE and time averaged RMSE for all configurations.

Baseline ($A_{\text{ex}} = 10, R = 0.5$)		
Method	Param ESS	State ESS
SIR	53 ± 5	17 ± 2
OT	52 ± 8	17 ± 2
Low noise ($R = 0.1$)		
Method	Param ESS	State ESS
SIR	2 ± 1	1 ± 0
OT	2 ± 0	1 ± 0
Very low noise ($R = 0.0001$)		
Method	Param ESS	State ESS
SIR	1 ± 0	1 ± 0
OT	1 ± 0	1 ± 0
Balanced kernel ($A_{\text{ex}} = 6$)		
Method	Param ESS	State ESS
SIR	75 ± 2	24 ± 1
OT	71 ± 3	25 ± 1
Combined ($A_{\text{ex}} = 6, R = 0.1$)		
Method	Param ESS	State ESS
SIR	8 ± 3	2 ± 0
OT	8 ± 3	2 ± 0

Table 5: Mean and variance ESS for all configurations

producing wide uncertainty bands that do not shrink over time. This pattern was observed for both methods, suggesting that the issue is structural rather than method-specific. This suggests that the observation operator is weakly sensitive to f , making different values of the parameter difficult to distinguish from the available observations.

The combination of accurate state tracking, low state RMSE and divergence in f raises the question: what effect does f actually have on the system? To investigate this, all other parameters were

fixed to their baseline values while f was varied in the true state (see Figure 14). Varying f does not change the temporal frequency itself. Instead, it changes the timing of the frequency shift. As a result, at any given time point, many values of f may produce similar local state. For some runs, f could converge to the true value until the frequency shift occurs, after which it diverges (see Figure 3). However, because it diverges in both positive and negative directions, the mean f can still appear to converge.

Despite the poor identifiability of f , the estimated state remained accurate. This is surprising, especially since A and ν are often well estimated, leaving no parameter to compensate for errors in f . One possible explanation is that the observations contain sufficient information to accurately recover the evolving state while remaining relatively insensitive to variations in f . Although this hypothesis is consistent with the experimental results, further research is required to verify it.

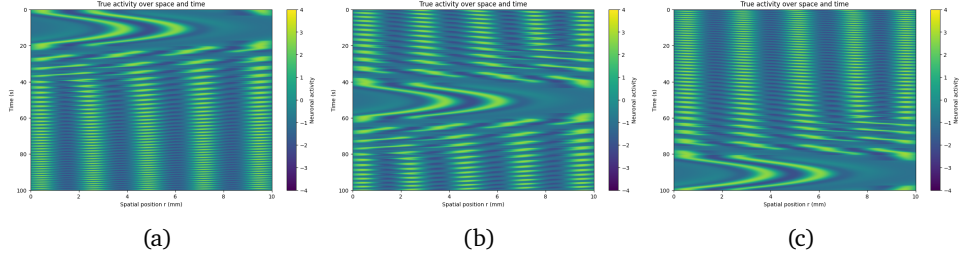


Figure 14: True activity state for $f = 0.1$ (a), $f = 0.5$ (b) and $f = 0.9$ (c). Varying f changes the timing of the frequency shift, but does not change the temporal frequency itself.

Since f is part of the phase term $\varphi(t) = f + at$, one might consider estimating a alongside θ . However, preliminary results show that even though f becomes slightly more stable, a oscillates and is not estimated accurately (see Figure 15). This behaviour is consistent with the ill-posed nature of inverse problems associated with the Amari neural field model. The inverse problem of reconstructing the synaptic connectivity kernel from observed neural field dynamics has been shown to be generally ill-posed, with solutions that are non-unique and highly sensitive to perturbations [Potthast and Graben, 2009]. Although their analysis focuses on kernel reconstruction rather than parameter estimation, the instability observed here suggests that estimating multiple interacting parameters in the Amari model may likewise suffer from poor identifiability.

Several limitations should be noted. First, the spatial resolution was fixed at $J = 25$. SIR may perform worse when the spatial dimension is increased due to curse of dimensionality [Reich and Cotter, 2015]. Although higher spatial resolutions were explored in the initial stages, this direction was not pursued further because of practical implementations constraints. Second, the Amari model does not produce chaotic dynamics [Coombes et al., 2014, Amari, 1977]. Consequently, the comparison presented in this thesis is limited to a relatively simple dynamical regime.

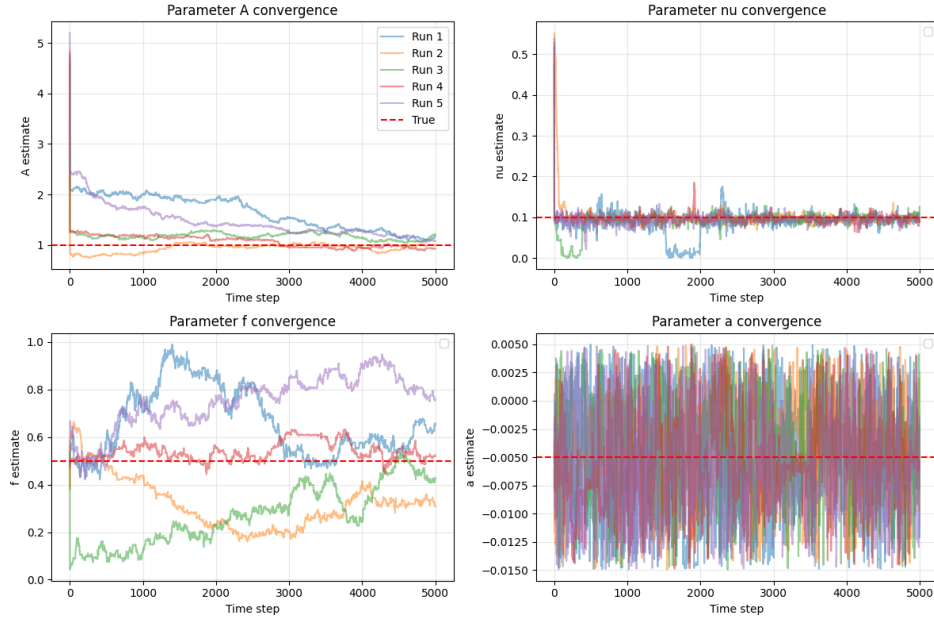


Figure 15: Preliminary results of estimating parameter a alongside θ .

For future work, several directions could be explored. One possible direction is to investigate whether the non-identifiability of f is specific to filtering methods. Since smoothing algorithms use the complete observation sequence rather than only past observations, they may better distinguish parameters whose influence becomes apparent only over longer time horizons. Another direction is to study how changes in the model parameters affect the difficulty of the joint estimation problem. For example, varying the connectivity length scales l_{ex} and l_{in} , which determine the spatial extent of excitatory and inhibitory interactions, may alter the underlying neural dynamics and, consequently, the performance of the filtering methods. Finally, extending the comparison to neural field models with more complex, potentially chaotic dynamics would help determine whether the conclusions drawn here generalise beyond the Amari model.

5 Conclusion

This thesis investigated joint state and parameter estimation in an Amari neural field model using Bayesian particle filtering methods. In particular, nested Sequential Importance Resampling (SIR) and nested transform-based particle filtering in the form of optimal transport (OT) were compared to determine whether transform-based particle filtering improves accuracy for both the hidden neural state and the unknown model parameters.

Across all experimental configurations, both methods achieve comparable state reconstruction errors and parameter estimation accuracy, with only minor regime-dependent differences. This contrasts with the expectation that transform-based methods may improve estimation accuracy.

An additional finding of this work was the persistent non-identifiability of the initial temporal frequency f . The estimates of f frequently diverged, despite appearing to converge on average. Preliminary experiments further suggested that jointly estimating the factor a does not resolve this issue, indicating that the underlying inverse problem may be inherently difficult for these parameters.

Overall, our results suggest that, for the Amari model under the tested conditions, nested SIR is sufficient for joint state and parameter estimation. The additional complexity of OT does not provide a clear practical advantage.

Popular Summary

The brain is one of the most complex systems in nature. It contains billions of nerve cells that constantly communicate with each other, producing rich and changing patterns of activity. These patterns depend on what the brain is doing, such as whether we are awake, asleep, or focused on a task.

To study this kind of activity, scientists use mathematical models. In this thesis, we focus on neural field models, which describe brain activity on a large scale rather than tracking individual nerve cells. These models can produce a variety of patterns depending on their settings.

A key difficulty is that we cannot directly observe either the true brain activity or the exact settings. Instead, we only have access to limited and noisy measurements. Because of this, both the true activity and the model parameters must be estimated from data.

To do this, we use particle filtering methods. A standard approach is Sequential Importance Resampling (SIR). A newer alternative is based on optimal transport, a technique that moves and reorganises information in a more structured way. This method has shown good performance in other fields, and the goal of this thesis was to test whether it could improve results when applied to a neural field model.

This work contributes to a broader effort to develop reliable methods for interpreting brain measurements. The results show that both methods perform similarly in terms of accuracy. However, the standard method is computationally cheaper and easier to implement, making it the more practical choice in this setting.

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